

PFMEA Case Studies - Real World Automotive Examples

Lessons Learned | Quality Head Agents

Case 1: Weld Defect Escape - Tier 1 Supplier

Problem: Incomplete weld on structural bracket reached customer assembly line - line stoppage. PFMEA had S=9, O=3, D=3 but action was "monitor during audit" (insufficient). Root Cause: Detection control relied on visual inspection - not reliable for sub-surface defects. Lesson: For S=9-10, detection must be poka-yoke, automated sensor, or 100% NDT. Changed to 100% UT inspection with recorded result. AP moved from M to H - immediate action implemented.

Case 2: Dimension Drift Not Detected - CNC Machining

Problem: Critical bore diameter drifted over 3 shifts due to tool wear. 450 parts shipped out of tolerance. PFMEA had D=4 (1st-off + last-off only). Root Cause: Sampling plan insufficient for tool wear rate. Lesson: SPC frequency must match process capability and tool life. Implemented in-process gauging every 25th part + SPC chart with control limits. Cpk improved from 0.98 to 1.67.

Case 3: PFMEA Not Updated After Engineering Change

Problem: Customer noise complaint on new model. Investigation found PFMEA was not updated when a component specification changed. Old detection controls were insufficient. Root Cause: No ECN process linked to PFMEA review. IATF Major NC raised (Clause 8.5.6.1). Lesson: Every ECN must trigger PFMEA review. Implemented: ECN triggered PFMEA review within 5 working days - mandatory.

Case 4: High AP Item Left Open - IATF Finding

Problem: IATF audit found 7 High AP action items from 4 months prior - all "In Progress" with no evidence. Major NC: Clause 8.5.6.1. Lesson: High AP items must have: Owner, Target Date, Evidence of Action, Verification of Effectiveness. Implemented monthly PFMEA action review meeting with Plant Head for all H-AP items.